

Intern Report

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1 Abstract

We aim to utilize various kinds of machine learning techniques to help us decide the strategies to use in the stock market, including some basic time-series models, the DRL method, and Large Language Models such as LLaMa3 and chatgpt4. In all the experiments, we consider some authoritative stocks in Taiwan such as 0050, 2330, 2454, etc. For most the experiments, the baselines are the strategy of buy-and-hold and some existing strategy based on some indices like RSI, K, D, MACD, etc. TimesNet [1] was used in our experiments.

2 Experiments

2.1 Time-series Predictions

2.1.1 Experiment Descriptions

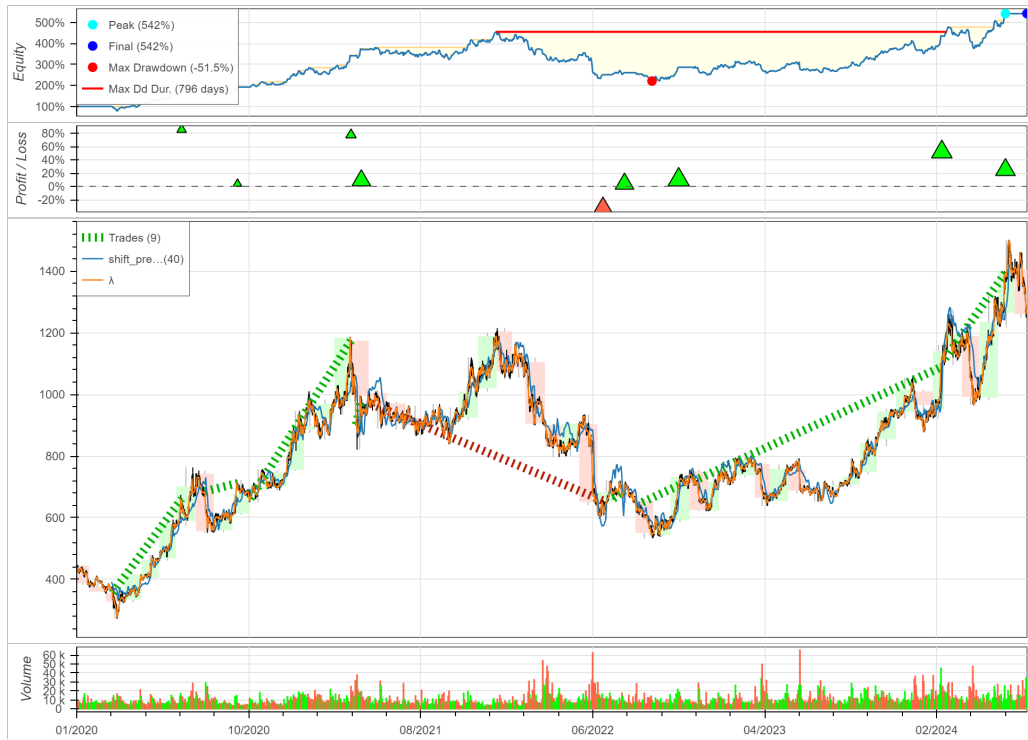
The most intuitive way of aiding stock trading with Machine Learning is to directly predict the future price. Therefore the first attempt was to use a simple CNN model to predict the price of the following weeks, and the proposed trading strategy was simple as well: Buying shares when the predicted price is higher than the current price, and vice versa. As expected, the results of the first experiment failed to outperform the strategy of buy-and-hold in terms of final returns due to the oversimplified model architecture and lack of diversity in input features. Hence, it was replaced by some state-of-the-art Time-series forecasting model, such as ITransformers and TimesNet. Moreover, a variety of key input features were fed into these models such as MACD signal, RSI index, the trading volume of foreign investors and investment trusts so as to equip the model with more information about the dynamic stock market. Besides, instead of buying and selling shares once the predicted price is higher or lower, a more reserved approach was utilized: transactions proceed only when the difference in share prices reaches a fixed threshold. The results of the experiment with these revised factors turned out to be quite encouraging, as we could see a substantial growth in overall performances. Presented as follows are the final results of our experiments which successfully outstripped most of the baseline strategies.

2.1.2 Experiment Settings

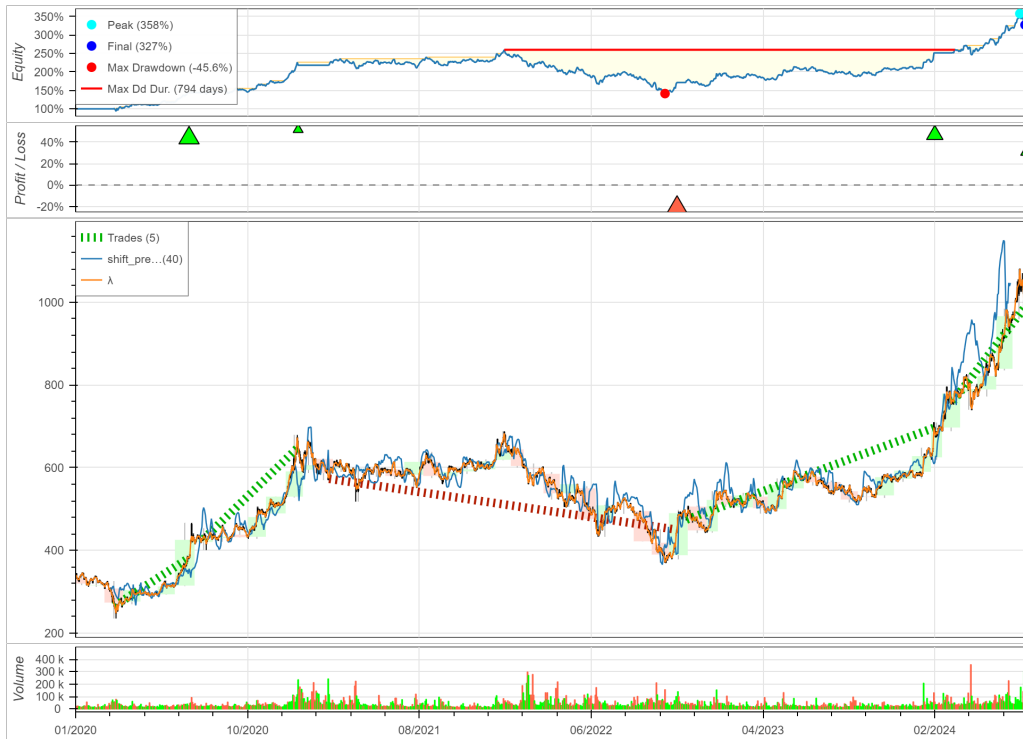
- Model Input Features: opening price, closing price, highest price, lowest price, number of shares traded, trading value, RSI, MACD, MACD-signal, K, D, the buying and selling volume of foreign investors and investment trusts (separately) over the last 40 business days
- Model Output: the predictions of the closing price of the next 20 business days
- Training Period: 2008-01-01 to 2019-12-31
- Validation Period: 2020-01-01 to 2024-07-22
- Model: TimesNet

2.1.3 Experiment Results

We present the results of the proposed method compared to some baseline strategies in Table 1 to quantitatively assess our work. Besides, we show some of the backtesting results in Figure 1 to 2.

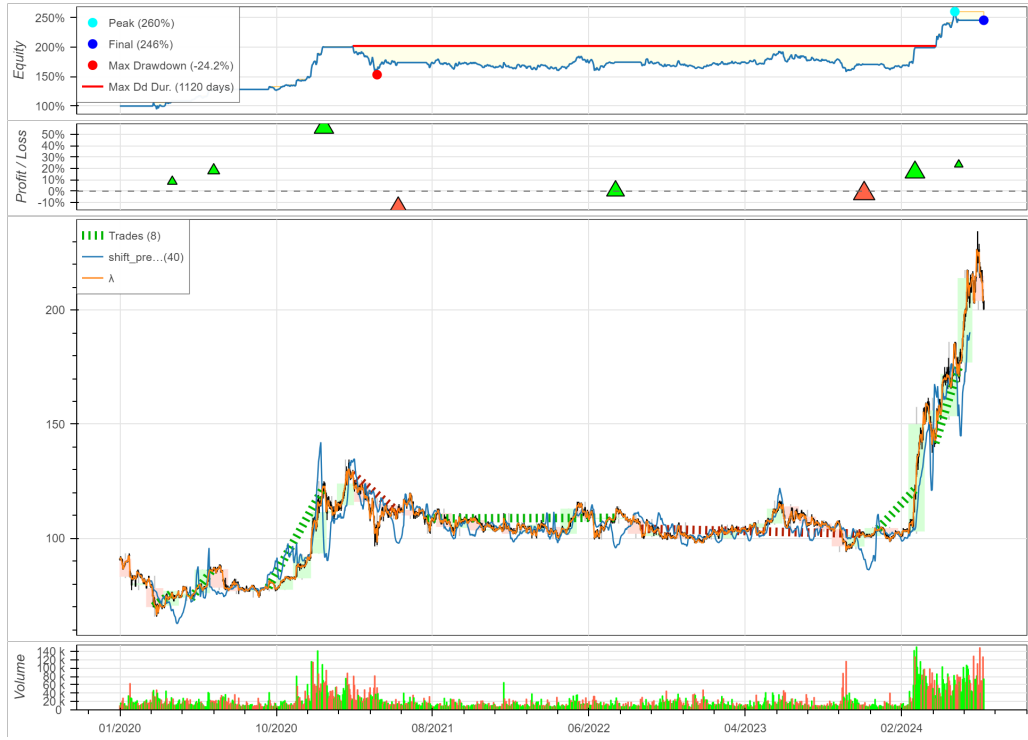


(a) 2454,5,5,-10

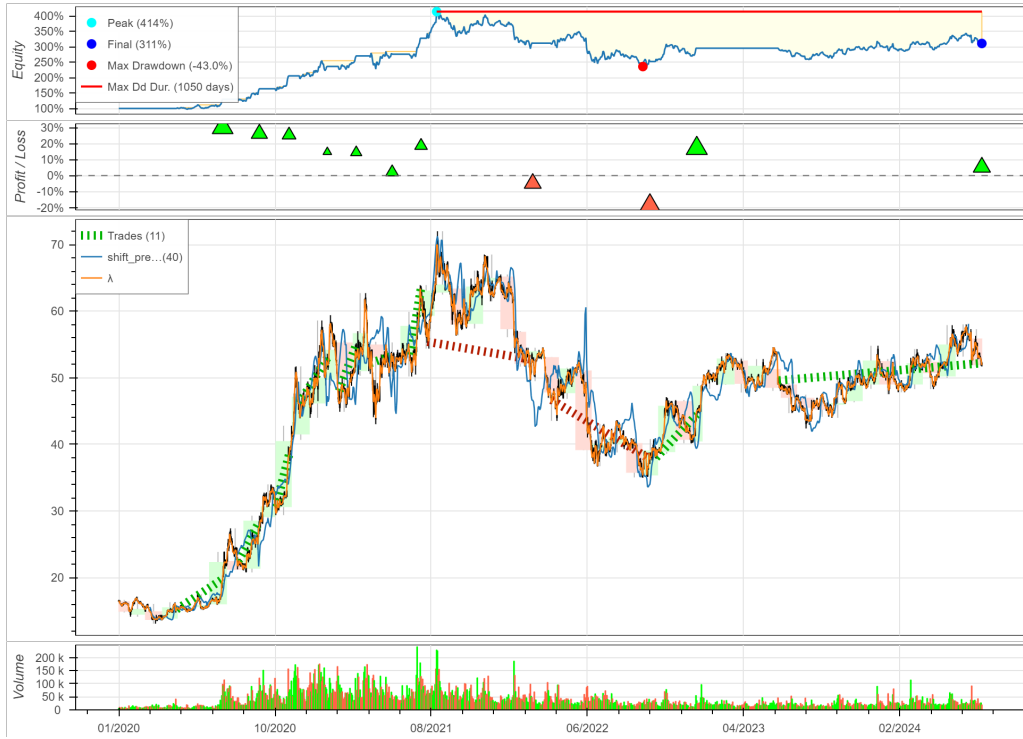


(b) 2330,20,10,5,-10

Figure 1: Backtesting Results(1)



(a) 2317,20,5,-10



(b) 2303,20,5,-10

Figure 2: Backtesting Results(2)



(a) 3228,20,5,-10



(b) 3228,MACD

Figure 3: Backtesting Results(3)

	Buy and Hold	MACD	20,5%,-10%	10,5%,-10%	20,10%,-10%	5,5%,-10%
2330	286%	259%	281%	302%	327%	387%
2454	285%	209%	321%	341%	258%	542%
0050	191%	195%	185%	192%	90%	228%
3228	938%	2295%	346%	487%	486%	320%
2317	224%	137%	246%	300%	218%	251%
2303	314%	197%	311%	304%	273%	257%
2891	160%	135%	195%	217%	115%	200%

Table 1: The above is the results of the strategies utilizing the predictions made by the TimesNet model compared to 2 baseline strategies: Buy-and-Hold and MACD. In the first of each column(except baselines), there are 3 numbers: n,buy%,-sell%, this represent the trading strategy defined as follows:

- Buy Signal: when the predicted price of n days later is buy% higher than the current price.
- Sell Signal: when the predicted price of n days later is sell% lower than the current price.

2.1.4 Discussion

As we can see from the result of Table 1, our proposed method have succeeded in beating the 2 baseline strategies(other baseline strategies like RSI or KD often perform poorer than MACD and Buy and Hold in the long run and thus are not explicitly showed in our table) Specifically, our approach was eclipsed by baselines strategies only for stock 3228 and 2303, and for the latter our approach was actually only beat by 3% in terms of final return. We can try to qualitatively list some of the backtesting results shown in Figure 1 to Figures 3 to gain more insight into our approach:

1. Figure 1: we showcase 2 stocks, 2330 and 2454, where our approach performs the best compared to the baseline strategies. We can find there are some common traits in the price of the 2 stocks: First, there are both multiple periods of continual price-growing trends that the model successfully foresaw. Second, there aren't too many short term periodic oscillation in price.
2. Figure 2: There 2 specific settings where our approach failed to beat the baselines for different reasons : In Figure 2(a), we can find that the fluctuation of price remains within a limited range for up to roughly 2 years , which means during the period, there's little room to utilize predictions to make profit. While in Figure 2(b), there's intense short-term fluctuation in price for a considerable amount of time. The haphazard behavior of price might account for the lackluster performance of our approach during the period.
3. Figure 3: For stock 3228, we can see that all our approaches pale in comparison to the 2 baseline strategies. A reasonable explanation might be that we accidentally select a point when the stock price reaches all-time low, which amounts to the dominant performance of the buy-and-hold strategy. Besides, in comparison to most stocks whose price has an inclination to keep going up in the long run, this specific stock might be inherently different because the expectation of stock price seems to remain within a limited range.(This observation is made from historical data.)

2.2 DRL Agent for Strategies Selections

2.2.1 Experiment Descriptions

The other attempt was to train the model as an agent, that is, instead of giving predictions on the stock price, the model would directly output the whether we should buy, hold, or sell. The benefit of this approach was that it realized an end-to-end service to the client without the need to devise a rigid trading strategy like we did in previous experiments. To implement this notion, we would have to utilize the technique of Deep Reinforcement Learning. More Precisely, the agent decided whether to buy, hold, or sell, at the start of each period by a policy model, which would predict the reward of each action given the environment features. After iterating a whole episode, we will update the agent based on the total reward, which can be spilt into 2 parts:

1. Long-Term Reward: The return of the whole episode multiplied by a discount factor
2. Short-Term Reward: The return of that specific period

And the return is defined the growth in equity in percentage. With this framework, we anticipated the agent to learn a more flexible trading strategies via the diversified features provided. To our frustration, the outcome turned out to be quite the opposite, the agent’s strategy had a strong tendency to converge to the strategy of buy and hold, which was actually quite reasonable considering under most circumstances the stock price always increase in the long term. We have tried multiple different settings, including the length of a period and the weight in each reward, but the effort turned out to be futile as we encountered the same result over and over again. Hence, we deduced that we need to adopt a different approach to tackle the problem. Instead of giving the model complete freedom to select from buy, hold, or sell during a relatively short period(say, a week), we modify the action space to be choosing from some existing strategy , such as buy and hold, MACD, KD, etc, and exert that strategy for a longer period of time(a month). By basing the actions on existing strategies that have already been proved to be effective in stock market, we hope that the model can perform better than the baselines. Fortunately, the outcome turned out to be much more valuable, as we can see the agent to utilize all sorts strategy make profit over the time span.

2.2.2 Experiment Settings

- Model Input Features: opening price, closing price, highest price, lowest price, number of shares traded, trading value, RSI, MACD, MACD-signal, K, D, the buying and selling volume of foreign investors and investment trusts (separately) over the last 40 business days
- Model Output: the predicted reward of all actions(strategies) for the next period.(then choose the maximum)
- Training Period: 2008-01-01 to 2019-12-31
- Validation Period: 2020-01-01 to 2024-07-22
- period length: 20 business days
- Model: TimesNet
- Stocks: 2330, 2454, 2308, 2891, 2317, 0050
- Strategies(Actions):
 1. RSI Strategy:
 - Uses 14-day Relative Strength Index (RSI)
 - Buy Siganl: when RSI crosses above 30 (oversold)
 - Sell Signal: when RSI crosses above 70 (overbought)
 2. SMA Cross Strategy:
 - Uses 10-day and 20-day Simple Moving Averages (SMA)
 - Buy Signal: when 10-day SMA crosses above 20-day SMA
 - Sell Signal: when 10-day SMA crosses below 20-day SMA
 3. KD Strategy (Stochastic Oscillator):
 - Uses 14-day window and 3-day smoothing
 - Buy Signal: when %K crosses above %D and %K is below 20 (oversold)
 - Sell Signal: when %K crosses below %D and %K is above 80 (overbought)
 4. MACD Strategy:
 - Uses 12-day fast EMA, 26-day slow EMA, and 9-day signal line
 - Buy Siganl: when MACD line crosses above signal line
 - Sell Signal: when MACD line crosses below signal line
 5. Buy and Hold Strategy:
 - Buy the entire portfolio if no current position
 - Sell at the end of the period

	Buy and Hold	MACD	Moving Average	KD	RSI	DRL Agent
2330	277%	259%	184%	87%	102%	286%
2454	285%	209%	285%	119%	128%	369%
0050	193%	202%	199%	102%	110%	165%
2317	212%	140%	140%	149%	159%	183%
2308	247%	199%	202%	125%	133%	288%
2891	159%	135%	131%	112%	115%	141%

Table 2: The above table show the results of the proposed DRL agent compared to the strategies in its action space. Because we have add some randomness to the model output, the presented number is actually the average over 30 data.

2.2.3 Experiment Results

We provide the results of our DRL agent accompanied by the performance of some baseline strategies in Table2. And we present some of our agents backtesting result in Figure 4

2.2.4 Discussion

As we can see, our agent successfully select the suitable combination of strategies to outshine the baseline strategies. The logic behind this approach is a to trade the degree of freedom of the agent for the rudimentary knowledge about the stock market. With these already widely-used strategies, we can preclude the model from making obvious and unreasonable blunder in stock market such as not buying anything throughout the whole process(which was encountered multiple times in the previous settings.) However, there’s a potential risk: the randomness added to the model resulted in a relatively large standard deviation in the data(might be up to over 100%). We tried to address this issue in the next experiment.

2.3 DRL Agent of Stock Selection

2.3.1 Experiment Descriptions

In light of the optimistic result of the last experiment, we continued on the setting and tried to resolve the risk of large randomness. Our approach was to devise another model particularly for stock selection in a similar manner. Features of multiple stocks were fed into the model, and the model would output the 2 stocks which it predicted to have the top 2 return in that period. After the model decide the stocks, the agent would use all its capital available capital to invest in these 2 stocks, and the strategy to use will be decided by the model of the previous experiment which had already been trained in advance.

2.3.2 Experiment Settings

- Model Input Features: opening price, closing price, highest price, lowest price, number of shares traded, trading value, RSI, MACD, MACD-signal, K, D, the buying and selling volume of foreign investors and investment trusts (separately) for all stocks considered over the last 40 business days
- Model Output: the predicted reward of all actions(stocks) for the next period.(then choose the top 2)
- Training Period: 2008-01-01 to 2019-12-31
- Validation Period: 2020-01-01 to 2024-07-22
- period length: 20 business days
- Model: TimesNet
- Stocks to choose(action space): 2330, 2454, 2308, 2891, 2317, 0050



Figure 4: DRL trading strategy results for stock 2330



Figure 5: DRL trading strategy results for stock 2454

stocks	2454	2308	2330	0050	2891	2317	RL-agent
Buy and Hold	277%	247%	277%	193%	159%	212%	288%(avg of last 30)
MACD	209%	199%	259%	202%	135%	140%	288%(avg of last 30)

- standard deviation: 52%

Table 3: The above table show the results of the proposed DRL agent for stock selection compared to the 2 baseline strategies. Because we have add some randomness to the model output, the presented number is actually the average over 30 data.

2.3.3 Experiment Results

The result is presented in Table 3 along with baselines. Because the backtesting package can only show the trading simulation for one stock, this we couldn’t explicitly show the diagram like the last 2 experiments.

2.3.4 Discussion

As we can see in Table 3, this experimental setup successfully reduces the standard deviation by a considerable amount. This experiment can been interpreted as a trade-off between the potential and the reliability. We sacrificed performance to a certain degree for the stability of the return over the whole period.

3 Conclusion

This study explored various machine learning techniques to develop strategies for stock market trading, focusing on prominent Taiwanese stocks. We investigated time-series models, Deep Reinforcement Learning (DRL) methods, and considered the potential of Large Language Models like LLaMa3 and ChatGPT4.

Our experiments yielded several key findings:

1. **Time-series Predictions:** Advanced models like TimesNet, when fed with diverse input features, outperformed baseline strategies such as buy-and-hold and MACD for most stocks. The approach was particularly effective for stocks with consistent growth trends and limited short-term fluctuations.
2. **DRL Agent for Strategy Selection:** We successfully trained an agent to select from existing trading strategies, overcoming initial challenges where the model tended to converge to a simple buy-and-hold strategy. This approach demonstrated the potential of combining machine learning with established trading methods.
3. **DRL Agent for Stock Selection:** Building on the strategy selection model, we developed an agent for stock selection. This approach reduced the standard deviation of returns, offering a balance between potential returns and reliability.

Throughout our experiments, we consistently outperformed baseline strategies for most stocks, demonstrating the potential of machine learning in stock market analysis and decision-making. However, we also observed limitations, particularly in stocks with unique characteristics or during periods of intense short-term fluctuations.

Future work could explore the integration of Large Language Models, which were not fully utilized in this study. Additionally, further refinement of the DRL models, expansion of the stock selection pool, and incorporation of more diverse features could potentially enhance the robustness and applicability of our approach.

In conclusion, while our machine learning-based strategies show promise, they should be viewed as tools to augment rather than replace human decision-making in stock trading. The complex and dynamic nature of financial markets necessitates a cautious approach, combining algorithmic insights with human expertise and risk management strategies.

References

- [1] Haixu Wu et al. “TimesNet: Temporal 2D-Variation Modeling for General Time Series Analysis”. In: *arXiv preprint arXiv:2210.02186* (2022). URL: <https://arxiv.org/abs/2210.02186>.